Aryan Ballani

Software Development, Distributed Systems, and AI Infrastructure

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Education

BSc. Computer Science and Statistics | University of British Columbia *Graduation: Apr 2026*

- **Dean's Honor List** for 3 consecutive years; GPA: 4.0/4.33.
- Awarded the **International Student Scholarship** for academic excellence and community involvement.

Work Experience

Machine Learning Engineer — *Railtown AI Technologies Inc., Vancouver, BC* *Jan 2025 – Present*

- Core contributor to Railtracks, an open-source Python agentic framework; shipped **38 merged PRs** and contributed **230+ commits** across a 5+ engineer team.
- Led a 3-phase backend migration from LlamaIndex to LiteLLM across **47 files and 1,300+ lines**, preserving feature parity with **100% CI pass rate and zero breaking changes**.
- Reorganized **5,100+ lines across 125 test files**, raising coverage to **91%** and reducing unit-test runtime from 76s to 7.6s; helped scale the suite beyond **2,200 tests**.
- Built production agent infrastructure spanning **async execution**, **real-time streaming**, REST APIs, GitHub webhooks, Dockerized benchmarking, and LLM evaluation.
- Designed **human-in-the-loop middleware** with Verdict semantics and timeout safety, backed by **12 tests**.
- Built and deployed a production **RAG documentation agent** in **2 weeks** with FastAPI, Streamlit, Postgres/pgvector, release webhooks, and automated re-indexing.

Software Engineer in Test — *BC Liquor Distribution Branch, Burnaby, BC* *May 2024 – Dec 2024*

- Automated **80% of manual test cases** using Selenium and Oracle SQL, increasing QA throughput by 25%.
- Profiled and optimized **30+ database queries** and integrated **100+ unit tests**, improving application performance by 12%.
- Integrated CI/CD workflows with Azure DevOps, reducing release-cycle time by **30%** across five QA modules.

Undergraduate Teaching Assistant — *UBC Computer Science Dept., Vancouver, BC* *Jan 2024 – Present*

- Led weekly labs for 50+ students covering programming fundamentals, data structures, C/C++, pointers, memory management, and systems debugging; provided code reviews and technical mentorship.

Relevant Projects

Resolution — Go, Gin, React, PostgreSQL, WebSockets, AWS ALB *Apr 2026*

- Designed a distributed real-time workspace using a block-level CRDT and **Hybrid Logical Clock ordering**, enabling concurrent edits to converge without stale writes.
- Built Go/Gin REST and WebSocket services with PostgreSQL persistence, optimistic conflict handling, shard-aware routing, and a **10+ block** editor; deployed on AWS behind an **Application Load Balancer**.

GlideCart — Python, OpenCV, YOLOv8, TensorFlow Lite, Raspberry Pi, Android, Supabase *Jan 2026*

- Built an autonomous shopping-cart prototype with **YOLOv8 object detection**, **OpenCV tracking**, and **TensorFlow Lite edge inference** on a Raspberry Pi 4, enabling real-time operation without cloud dependency.
- Developed an **Android companion app** and **Supabase/PostgreSQL synchronization layer** for live camera feeds, checklist events, and manual controls across device and mobile clients.

Campus Capture — TypeScript, JavaScript, HTML, CSS, Mocha, Chai *Jan 2024*

- Built modular RESTful endpoints and an asynchronous backend for adding, querying, and deleting **20,000+ course datasets**; implemented logical query operators across 10+ fields.
- Added defensive validation, crash-resistant error handling, and **100+ automated tests**, maintaining sub-500ms responses across malformed-input and edge-case scenarios.

Technical Skills

Languages	: Python, C++, C, Java, Go, TypeScript, JavaScript, SQL
Software	: Data Structures, Algorithms, OOP, Async Programming, REST APIs, WebSockets
Systems	: Distributed Systems, Networking, PostgreSQL, MongoDB, Docker, AWS, Azure
Frameworks	: FastAPI, Flask, Gin, React, Vue.js, Express.js
Testing/AI Tools	: Pytest, JUnit, Selenium, Mocha, Chai, GitHub Actions, Azure DevOps,